

TECHNICAL PROPOSAL AND SOFTWARE REQUIREMENTS AND DESIGN DOCUMENT

Hadia Gifting Registry

Gift registry, item reservation and M-Pesa cash gifting platform with automated escrow and payouts.

PREPARED FOR

Hadia
Attn: Tracey, Founder

PREPARED BY

Codevertex Africa Limited
www.codevertexafrica.com

DATE

6 October 2026

DELIVERY PERIOD

12 weeks

INVESTMENT

KES 240,000 build
KES 8,000 per month

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Document control

Version	Date	Author	Description
1.0	6 October 2026	Codevertex Africa Limited	Issued for client review

Role	Party	Responsibility
Product owner	Hadia	Scope approval, business decisions, legal sign-off, pilot events
Delivery lead	Codevertex	Planning, scope control, reporting, risk management
Technical lead	Codevertex	Architecture, code quality, security, releases

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Executive summary

Codevertex Africa Limited proposes to design, build and launch the Hadia Gifting Registry, a mobile-first platform on which registry owners list gifts for an occasion, share a single link, and receive item reservations and M-Pesa cash contributions from guests. Cash contributions are held in escrow and paid out to owners automatically, net of Hadia's commission, under rules configured by Hadia.

The platform is built on Codevertex's production services for identity, payments, accounting, escrow, approvals and messaging. The registry service, web application and administration console are developed specifically for Hadia.

Key outcomes

- Registries published and shared within five minutes of sign-up, with no application download for owners or guests.
- Duplicate gifts eliminated through atomic item reservations visible to all guests in real time.
- M-Pesa contributions credited exactly once and reconciled hourly against the PayHero wallet.
- Payouts and commission deductions executed automatically by threshold and event date, with approval required only above Hadia's limit.
- Commission transferred to Hadia's business account daily without manual intervention.
- Compliance with the Data Protection Act, consumer protection, payments and accessibility requirements from launch.

Engagement summary

12 weeks Delivery period	KES 240,000 Fixed build price	KES 8,000 Monthly service fee: treasury subscription and dedicated support	KES 12,000 Hosting per year, billed to Hadia
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SECTION 1

Introduction

1.1 Purpose

This document specifies the requirements, design, delivery plan and commercial terms for the Hadia Gifting Registry. It is the basis for development, testing and acceptance.

1.2 Product scope

Area	Included
Public website	Landing page, how it works, occasions, help centre, legal pages, cookie consent
Owner application	Mobile sign-up, registries, items, sharing, gift log, thank-you tracking, payout settings, premium upgrade, privacy settings
Guest experience	Registry page, reservations, M-Pesa contributions (Airtel Money and card where enabled), receipts, reservation cancellation
Funds management	Escrow per registry, automated payouts, commission deduction and sweep, contribution limits, refunds, reconciliation, accounting
Administration	Registry and user management, approvals, exception queue, reports, fee and limit settings, moderation, data requests
Messaging	SMS and WhatsApp receipts and alerts, email, marketing opt-in
Compliance	Legal policy drafts, consent records, data export and deletion, WCAG 2.2 AA accessibility

Excluded: native iOS and Android applications, wedding planning and RSVP tools, gift delivery and retail, retailer affiliate integrations, formal legal opinions, marketing campaigns.

1.3 Definitions

Term	Definition
Registry	An owner's list of items and cash funds for an event
Escrow account	The treasury record holding a registry's cash contributions
Payout	Transfer of a registry balance, net of commission, to the owner
Commission	Hadia's percentage fee on cash contributions, deducted at payout
Sweep	Scheduled transfer of earned commission to Hadia's business account
Premium	One-time paid upgrade per registry
STK push	M-Pesa payment prompt sent to the payer's phone
PayHero Team	Hadia's PayHero account with an isolated wallet
KYC, AML	Know Your Customer; anti-money laundering
ODPC, DPIA	Office of the Data Protection Commissioner; data protection impact assessment
eTIMS	KRA electronic Tax Invoice Management System

PWA	Progressive web application
UAT	User acceptance testing
RPO, RTO	Recovery point objective; recovery time objective

1.4 References

- Hadia landing page concept and Gift Wishlist and Cash-Gifting Platform build report, August 2026
- Codevertex platform documentation: treasury escrow, PayHero integration, settlement flows, authentication, notifications
- Data Protection Act, 2019 and Data Protection (General) Regulations, 2021
- Web Content Accessibility Guidelines 2.2; OWASP Application Security Verification Standard 4

SECTION 2

Overall description

2.1 Product perspective

Hadia operates as a tenant of the Codevertex platform. Two new applications, hadia-ui and hadia-api, provide the registry product. Identity, payments, ledger, escrow, approvals and messaging are provided by existing platform services, described in [Solution architecture](#).

2.2 Product functions

- Registry creation and management for weddings, baby showers, housewarmings, graduations and birthdays
- Item listing by manual entry or product link, with reserve, fund or combined modes
- Guest reservations without duplicates, and cash contributions by M-Pesa
- Escrow of contributions with automated payouts and commission deduction
- Premium upgrade per registry: custom link, printable list, instant payouts
- Administration, reporting, reconciliation and compliance management

2.3 User classes

Role	Description	Permissions	Authentication
Guest	Family, friends, diaspora relatives, colleagues	View registry, reserve, contribute, leave a message, cancel own reservation	None; OTP when reserving
Registry owner	Couple, parent, graduate or host, aged 18 or over	Manage registries and items, share, view gifts, thank guests, set payout number, purchase premium	Mobile number and OTP
Co-owner	Invited by the owner	Edit items, thank guests	Mobile number and OTP
Hadia support	Hadia staff	User support, moderation, data requests	SSO with MFA
Hadia finance	Hadia staff	Approvals, ledger, reports, fees and limits	SSO with MFA; OTP per approval
Hadia administrator	Founder and delegates	All permissions, staff roles, settings	SSO with MFA

2.4 Operating environment

- Guests and owners: any modern mobile or desktop browser; installable as a PWA
- Hosting: Codevertex Kubernetes platform with PostgreSQL, Redis, NATS JetStream and object storage, behind Cloudflare
- Payments: PayHero (M-Pesa, Airtel Money, card); messaging: Africa's Talking (SMS), Meta WhatsApp Cloud API, SMTP email

2.5 Constraints

- Funds are collected, held and disbursed only through a licensed payment service provider.
- The registry service does not move money; all payments are executed by the treasury service.

- Retailer product data is inconsistent; manual item entry is the primary method.
- Payment onboarding depends on Hadia's company registration and PayHero KYC.

2.6 Assumptions and dependencies

- Hadia is a registered Kenyan company with a KRA PIN and business bank account or paybill by week 2.
- Hadia completes PayHero KYC to tier 3 and provides requested documents within 3 working days.
- Hadia appoints a product owner with authority to decide within 2 working days.
- Hadia provides brand assets, approves copy and supplies consented testimonials.
- Hadia engages a Kenyan advocate to approve the legal pack and advise on payment regulation.
- Hadia recruits 3 to 5 pilot events for week 11.
- The treasury platform subscription, including escrow and M-Pesa features, and dedicated support are covered by the monthly service fee.
- Hadia bears hosting, domain, SMS, WhatsApp and payment provider charges as set out in the commercial terms.
- Hadia purchases and renews the domain name annually, or engages Codevertex to do so under the managed domain service.

SECTION 3

Functional requirements

Priority: **M** must have, **S** should have, **C** could have.

ID	Requirement	Pri.
ACCOUNTS AND ONBOARDING		
FR-01	Owners sign up and sign in with a mobile number and a 6-digit one-time password sent by SMS, with WhatsApp as fallback. Codes expire after 5 minutes. Sending is rate limited.	M
FR-02	Owners confirm they are 18 or over and accept the terms and privacy notice. Version and timestamp are recorded.	M
FR-03	Owners verify the payout M-Pesa number by one-time password. The registered M-Pesa name on the first payout is matched against the owner's name; mismatches are flagged.	M
FR-04	Owners invite co-owners by mobile number. Co-owners cannot change payout details.	S
REGISTRIES AND ITEMS		
FR-05	Create a registry with type, title, date, optional location, cover photo and description. An owner may hold multiple registries.	M
FR-06	Add items manually: title, description, price, quantity, photo, store name and mode (reserve, fund or both).	M
FR-07	Add items from a product link. Public page metadata prefills the form for owner confirmation.	S
FR-08	Create cash funds with an optional target.	M
FR-09	Reorder, edit, hide and archive items. Items with gifts can be archived but not deleted.	M
FR-10	Registry visibility: public, unlisted (default) or passcode protected.	S
SHARING AND GUEST EXPERIENCE		
FR-11	Each registry has a short link with a preview image, title and description for WhatsApp, Facebook and SMS.	M
FR-12	Guests view items, reservation status, fund progress and the event description without an account.	M
FR-13	Reservations are atomic; the last unit of an item cannot be reserved twice.	M
FR-14	Guests cancel their reservation from a link sent to them.	M
FR-15	Owners control whether reserving guests' names are visible to other guests. Hidden by default.	S
CONTRIBUTIONS AND PAYOUTS		
FR-16	Guests contribute toward an item or fund by M-Pesa STK push, within configured limits. Airtel Money and card are available where enabled.	M
FR-17	The registry receives the full gift. PayHero's 2.9% processing fee is shown before payment and added to the guest's total by default; Hadia can choose to absorb it instead.	M
FR-18	Where the STK push cannot be delivered, guests are offered a paybill and account number.	S
FR-19	Guests receive an SMS or WhatsApp receipt. Owners receive a gift notification.	M
FR-20	Payouts run automatically when the balance reaches the threshold and after the event date. Premium registries pay out on each contribution.	M

FR-21	Commission is deducted automatically at payout and itemised on every payout statement.	M
FR-22	Earned commission is transferred automatically to Hadia's business account on a schedule, above a minimum amount.	M
FR-23	Payouts at or above the approval limit require finance approval with OTP. Payouts below the limit run without intervention.	M
FR-24	Cancelling a registry refunds each contribution to the paying number. Contributions without a number are listed for manual refund.	M
FR-25	Owners download a statement of contributions and payouts in PDF or spreadsheet format.	S
OWNER TOOLS		
FR-26	Gift log with guest name, amount, item and message; thank-you status; WhatsApp thank-you message.	M
FR-27	Premium upgrade per registry: custom link, printable list, instant payouts.	M
FR-28	Reminders to guest and owner 14 days after an unpurchased reservation. Owner may release the item.	C
ADMINISTRATION		
FR-29	Search registries, owners and contributions. View a registry as a guest.	M
FR-30	Configure commission, premium price, contribution limits, payout thresholds, approval limit and sweep schedule.	M
FR-31	Exception queue: failed payouts, name mismatches, manual refunds, reconciliation variances, reported registries.	M
FR-32	Content moderation with recorded reasons and owner notification.	M
FR-33	Reports: contributions and value by period, active registries, commission, premium sales, conversion, payout times.	M
FR-34	Data subject requests (access, correction, deletion) with deadline tracking.	M
FINANCIAL CONTROLS		
FR-35	Card contributions are held for 14 days before becoming eligible for payout. Chargebacks and reversals are debited from the registry, then the commission reserve, then recovered from the owner.	M
FR-36	Scheduled promotional commission rates, including 0%, applied to registries created within the promotion window.	S
FR-37	Hadia can freeze and unfreeze payouts per registry or platform-wide, and trigger manual payouts.	M
FR-38	Owners raise payout disputes from the dashboard; disputes enter the exception queue.	S

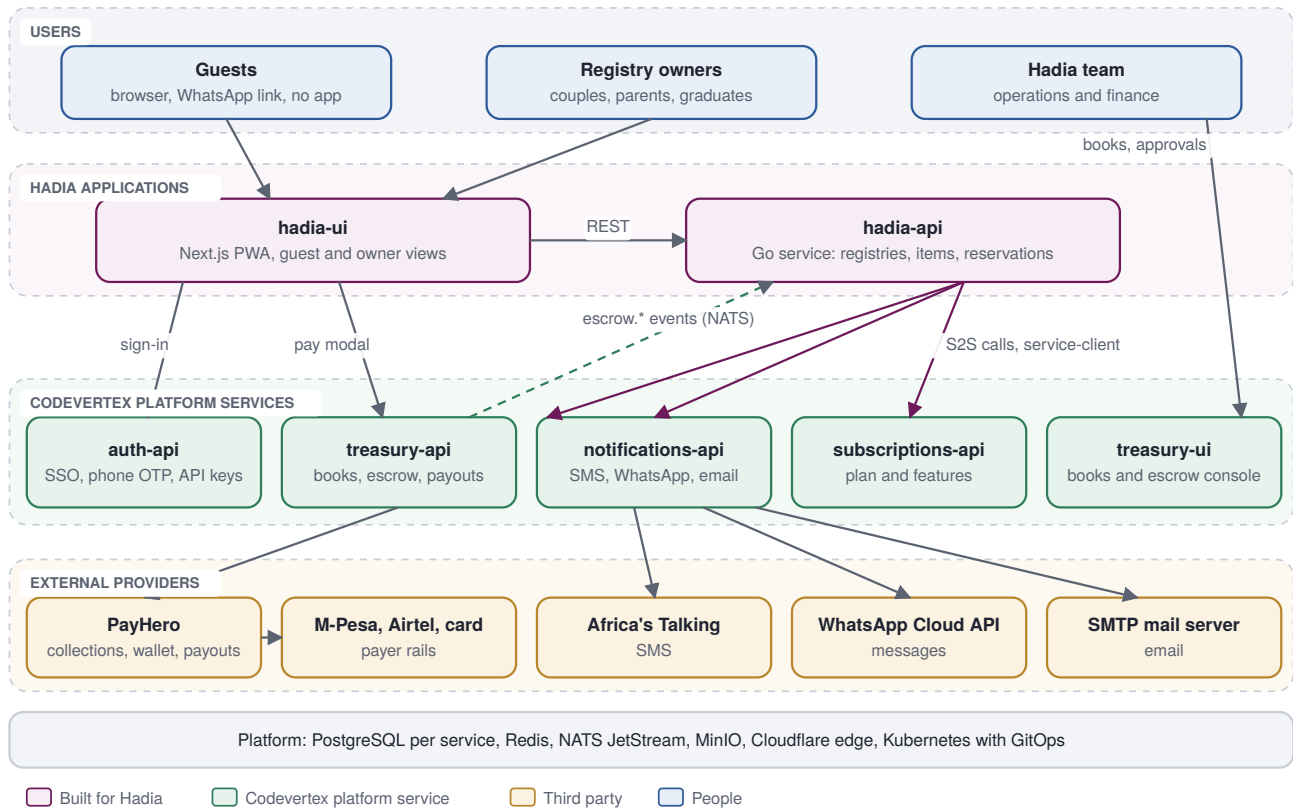
SECTION 4

Non-functional requirements

ID	Attribute	Requirement
NFR-01	Mobile performance	Guest page initial transfer under 150 KB excluding photos. Largest Contentful Paint under 2.5 seconds on a mid-range Android device over 3G. Photos served as resized WebP.
NFR-02	Response time	95th percentile under 300 ms for reads and 600 ms for writes, excluding payment provider latency.
NFR-03	Capacity	10,000 active registries and 2,000 contributions per hour at peak, scalable tenfold without redesign. Keyset pagination on all lists; reports served from daily aggregates.
NFR-04	Availability	99.5% monthly for guest pages and checkout. Minimum two replicas per service; zero-downtime deployments.
NFR-05	Financial integrity	Each contribution credited once, each payout executed at most once, balanced ledger, hourly reconciliation with the PayHero wallet.
NFR-06	Security	OWASP ASVS Level 2, TLS throughout, encrypted secrets, MFA for staff, audit trail on all financial and configuration changes.
NFR-07	Privacy	Data minimisation, consent records, data subject requests within statutory timelines, documented hosting location and transfer safeguards.
NFR-08	Accessibility	WCAG 2.2 Level AA.
NFR-09	Localisation	English at launch. Text externalised for Kiswahili.
NFR-10	Recovery	Daily encrypted offsite backups retained as 7 daily and 4 weekly copies; quarterly restore drills. RPO 24 hours, RTO 4 hours.
NFR-11	Observability	Structured logs, distributed tracing, alerts on failed payouts, reconciliation variances, error rates and latency.
NFR-12	Maintainability	Codevertex platform stack and conventions, automated tests, documented APIs.

SECTION 5

Solution architecture

**Figure 1.** Solution architecture

5.1 Components

Component	Status	Function
hadia-ui	New	Next.js progressive web app: website, guest pages, owner dashboard, administration console. Uses the Codevertex shared UI library for payments, sign-in, data tables, legal links and cookie consent.
hadia-api	New	Go service for registries, items, reservations, premium purchases, gift log, statistics and consent records. Does not move money.
auth-api	Existing, extended	Tenant, staff single sign-on with MFA, roles, terms acceptance, sessions. Extended with passwordless mobile sign-in for consumers.
treasury-api	Existing, extended	Ledger, PayHero gateway, checkout, escrow, approvals, payout dispatch, reconciliation, statements, eTIMS invoicing. Extended with scheduled commission sweep and per-registry contribution limits.
treasury-ui	Existing	Finance console: ledger, escrow accounts, approvals, wallet, reports.
notifications-api	Existing	SMS (Africa's Talking), WhatsApp (Meta Cloud API), email (SMTP), push. Hadia message templates added.
subscriptions-api	Existing	Hadia's platform plan and feature entitlements.

Shared libraries	Existing	Authentication client, service client with retries and circuit breakers, event outbox, cache and job leases, pagination, HTTP middleware, shared UI library.
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5.2 Architecture principles

- **Single data owner.** hadia-api references treasury records by ID. Balances are read from treasury.
- **Single payment path.** Only treasury-api communicates with PayHero.
- **Event driven.** Treasury publishes escrow events through a transactional outbox to NATS. hadia-api consumes them idempotently.
- **Replica safe.** Scheduled jobs run once per period across all replicas using shared leases.
- **Standard stack.** Go with Ent and Atlas migrations, Next.js, Helm and ArgoCD.

SECTION 6

Funds management and payouts

6.1 Custody

Hadia operates as a tenant on the Codevertex platform with its own PayHero Team account, registered in Hadia's name and verified to PayHero KYC tier 3. The Team wallet is isolated and used only by Hadia. All contributions are received into this wallet. Treasury maintains a separate sub-ledger for each registry, so the wallet balance is fully allocated between registries and Hadia's earned commission at all times.

Codevertex does not receive or hold Hadia funds or customer funds.

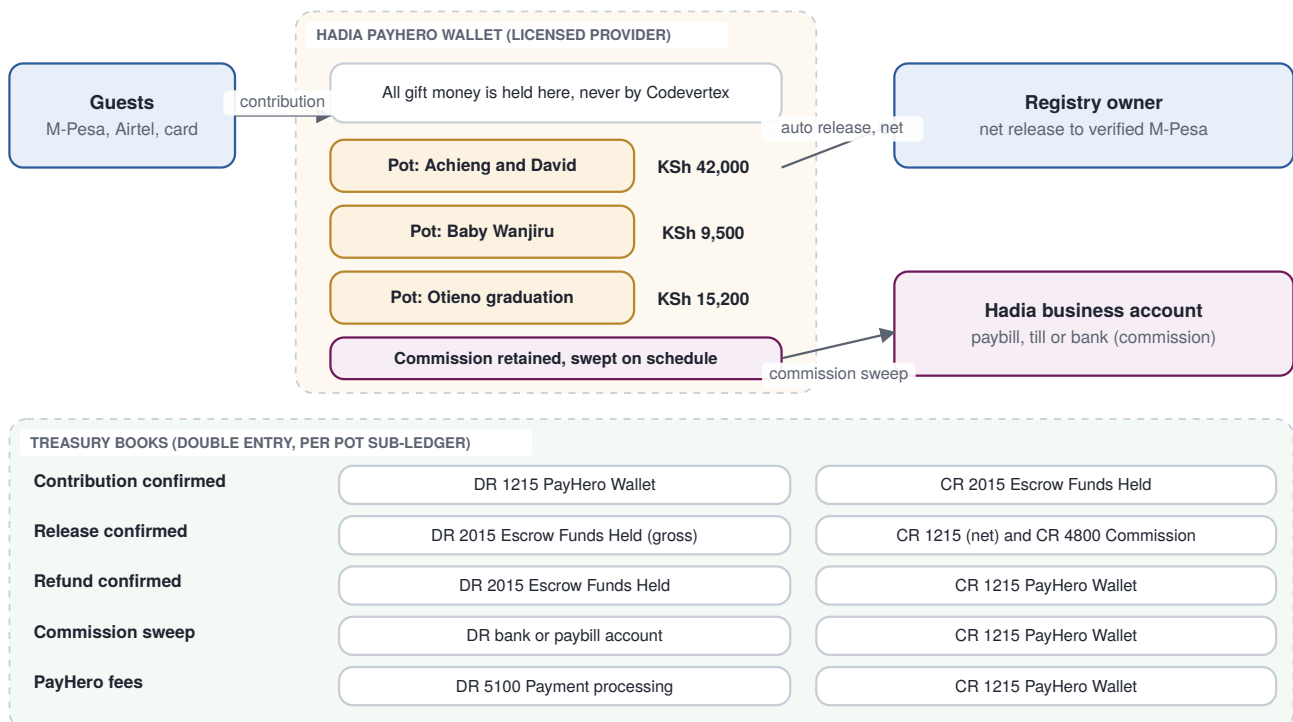


Figure 2. Funds flow and accounting entries

6.2 Registry escrow account

Each published registry has an escrow account in treasury. The account holds the registry title, the owner as beneficiary with a verified M-Pesa number, the target, payout rules and gift items. The registry ID is used as the external reference, which prevents duplicate accounts on retry.

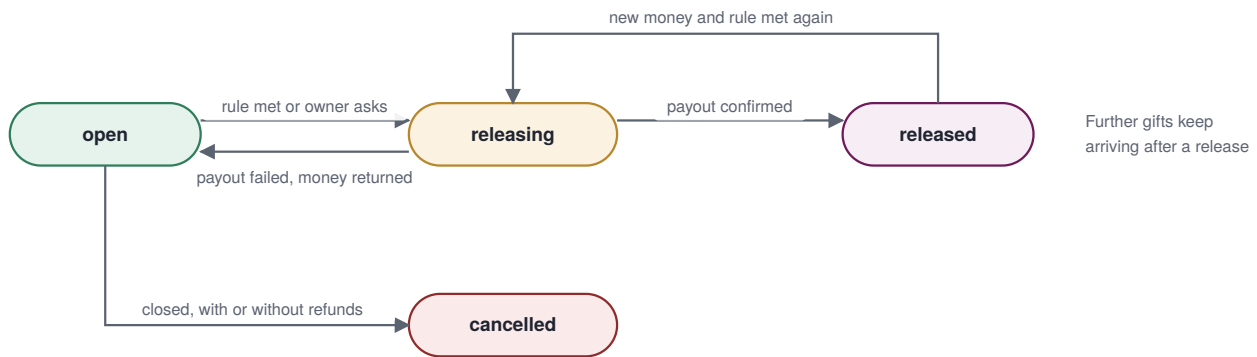


Figure 3. Escrow account lifecycle

6.3 Payout configuration

Setting	Level	Default	Behaviour
Automatic payout	Registry	On	Payouts run when a rule is met, checked every 5 minutes
Payout threshold	Registry	KES 10,000 standard; every contribution for premium	Full balance paid when the threshold is reached
Hold until	Registry	Event date plus 2 days	Remaining balance paid after this date
Approval limit	Hadia	KES 150,000 per payout	Payouts at or above the limit require finance approval with OTP, bound to amount and destination
Commission	Hadia, with registry override	4%	Deducted at payout and recorded as Hadia income
Contribution limits	Registry	KES 50 minimum, KES 150,000 maximum	Contributions outside the range are rejected before the STK push
Maximum balance	Platform	Agreed with Hadia	Contributions exceeding the cap are declined
Maximum hold	Platform	180 days	Funds cannot be held beyond this period
Commission sweep	Hadia	Daily above KES 5,000, KES 2,000 reserve	Earned commission transferred to Hadia's business account

6.4 Payout process

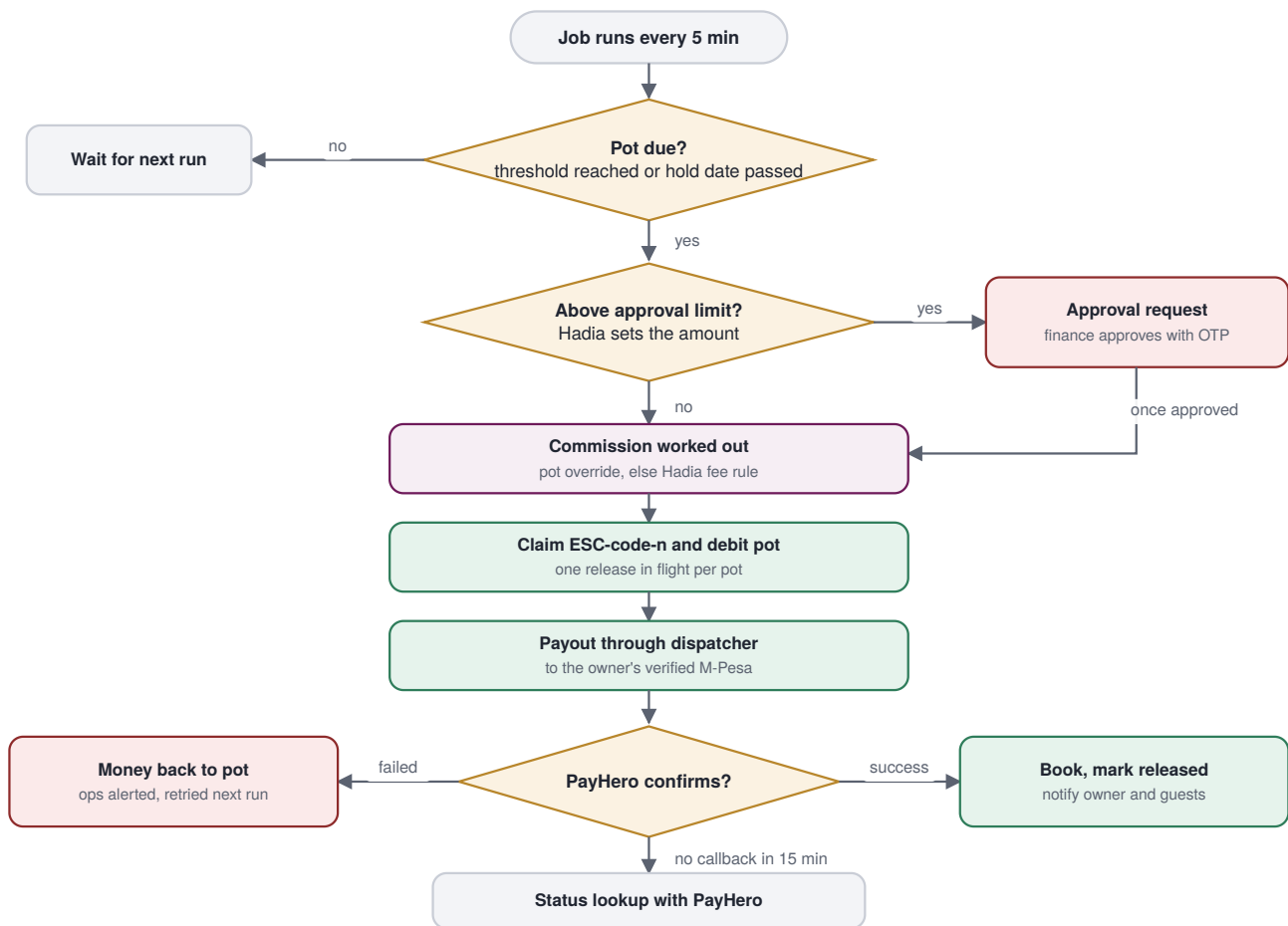


Figure 4. Automated payout process

Each payout attempt is registered under a unique reference before PayHero is called. Only one payout may be active per registry. A failed payout returns the funds to the registry balance. Payouts without a callback after 15 minutes are confirmed by status enquiry. All results are verified with PayHero before posting.

6.5 Commission

Commission is deducted from the gross payout and the net amount is paid to the owner. Commission is recorded in income account 4800 and retained in the Team wallet until swept.

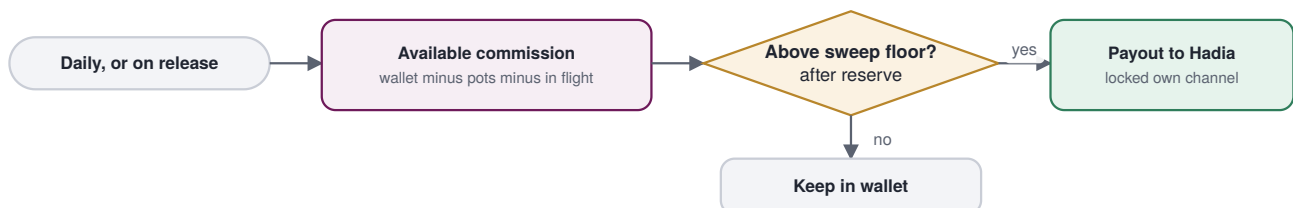


Figure 5. Commission sweep

1. The sweep runs daily, or after each payout if configured.
2. Available commission equals the wallet balance less registry balances, payouts in progress and the operating reserve.
3. Amounts above the sweep minimum are paid to Hadia's verified business channel. Changing the destination requires MFA and re-verification.
4. The transfer is posted from the PayHero wallet account to Hadia's bank or paybill account. The sweep is suspended if reconciliation reports a shortfall.

Manual withdrawal of commission remains available as an alternative.

6.6 Reconciliation

Treasury compares the total of registry balances and payouts in progress with the PayHero wallet balance every hour. Any shortfall is raised in the exception queue and suspends the commission sweep until resolved. Each registry statement is available in PDF, CSV and Excel.

6.7 Commission basis and configuration

Question	Position
Basis of calculation	Commission is charged on the full gift. PayHero's processing fee never reduces the registry balance.
Worked example	A guest gives KES 2,000 and pays KES 2,060, including PayHero's 2.9% processing fee. The registry is credited KES 2,000. At payout, 4% commission (KES 80) is deducted and the owner receives KES 1,920.
Processing fee	Added to the guest's total by default and shown before payment. If Hadia chooses to absorb it, the guest pays KES 2,000 and Hadia covers the KES 58 fee from its commission.
Payout fee	PayHero's payout fee is taken from the wallet and paid from Hadia's commission. The owner receives the full net amount.
Changing the rate	Hadia finance changes the commission rule (percentage, fixed amount, minimum and maximum fee) at any time from the administration console. Changes apply to payouts made after the change and are recorded in the audit log.
Per-registry rates	A different rate, including 0%, can be set on individual registries, for example for partner planners or VIP clients.
Promotional periods	Hadia can schedule a promotion with a start date, end date and rate, including 0%. Registries created within the window keep the promotional rate for their lifetime.

6.8 Payout controls

Control	How it works
Freeze a payout	Hadia can freeze any registry. Automatic payouts stop and the balance is held until the freeze is lifted. A platform-wide freeze switches every payout to manual approval.
Manual payout	Hadia staff, or the owner on request, can trigger an immediate payout of the available balance, subject to the approval limit.
Incorrect payout number	The payout number is verified by one-time password before the first payout, and the registered M-Pesa name is compared with the owner's name. A mismatch blocks the payout until reviewed. If funds still reach a wrong number, Hadia initiates a reversal through PayHero and Safaricom.

Payout dispute	The owner raises a dispute from the dashboard. The statement shows every contribution and payout with M-Pesa receipt numbers. Disputes enter the exception queue with a 2 working day response target.
Failed payout	Funds return to the registry balance automatically, the owner is notified, and the payout is retried on the next cycle. Repeated failures move to the exception queue.

6.9 Reversals, fraud and chargebacks

Event	Handling	Loss allocation
Payment reversed before payout	The reversal is debited from the registry balance and the owner is notified.	No loss
Payment reversed after payout	Recorded as an amount owed by the owner and recovered from future contributions to any of the owner's registries. Unrecovered amounts are pursued under the owner terms.	Hadia, pending recovery from the owner
Card chargeback	Card contributions become eligible for payout only after a 14-day hold. A chargeback is debited from the registry balance first, then from Hadia's commission reserve, then recovered from the owner. Hadia responds to the dispute with the contribution evidence held in the ledger.	Hadia, pending recovery
Fraudulent or unauthorised payment	Frozen on detection; refunded to the payer where confirmed. Monitoring flags unusual patterns such as many large gifts from few numbers.	Hadia as operator, except where caused by a Codevertex software defect

M-Pesa STK payments are authorised by the payer's PIN, so unauthorised M-Pesa contributions are rare. Card chargeback rights last up to 120 days under card scheme rules, which is why card-funded gifts carry the 14-day hold and a commission reserve.

6.10 Payment failure scenarios

Scenario	Guest experience	System behaviour
M-Pesa unavailable	Prompt fails; guest is offered the paybill option, Airtel Money or card	Contribution held as pending; no duplicate charge on retry
PayHero unavailable	Contributions temporarily unavailable; reservations continue	Payouts queue and resume automatically; funds remain safe in the wallet
Payout fails	Owner notified of the delay	Funds returned to the registry; retried next cycle
Registry cancelled	Guests notified that gifts are being returned	Each contribution refunded to the paying number

6.11 International guests

Question	Position
Visa and Mastercard	Supported through PayHero's hosted card checkout, subject to the card rail being enabled on Hadia's PayHero account at onboarding. Card data never touches Hadia systems.
Countries	Cards: guests worldwide, subject to issuer approval. Mobile money: the 14 countries listed below.
Currencies	Registries are held in Kenya shillings. Mobile money outside Kenya is charged in the local currency; cards are charged in KES.

Foreign exchange	Cards: the guest's card issuer converts at its rate. Mobile money: treasury converts at a stored exchange rate refreshed every 6 hours and rounded up; the conversion is recorded on each payment.
Fees to international guests	Hadia charges guests nothing. The guest's bank may apply its own foreign transaction fee. PayHero's card processing charge is borne by Hadia.

6.12 Supported mobile money countries

Region	Country	Charged in
East Africa	Kenya	KES
East Africa	Uganda	UGX
East Africa	Tanzania	TZS
East Africa	Rwanda	RWF
East Africa	Burundi	BIF
East Africa	Ethiopia	ETB
Central Africa	Democratic Republic of the Congo	CDF
Southern Africa	Zambia	ZMW
Southern Africa	Malawi	MWK
Southern Africa	Mozambique	MZN
Southern Africa	Botswana	BWP
Southern Africa	South Africa	ZAR
West Africa	Nigeria	NGN
West Africa	Ghana	GHS

Kenya is live at launch through M-Pesa and Airtel Money. Each other country is enabled when PayHero activates the mobile money rail for that country on Hadia's account; networks offered (for example MTN, Airtel) follow PayHero's coverage in each country.

SECTION 7

Process workflows

Each workflow defines its trigger, steps, automation and exception handling. Steps marked **AUTO** run without user or staff action.

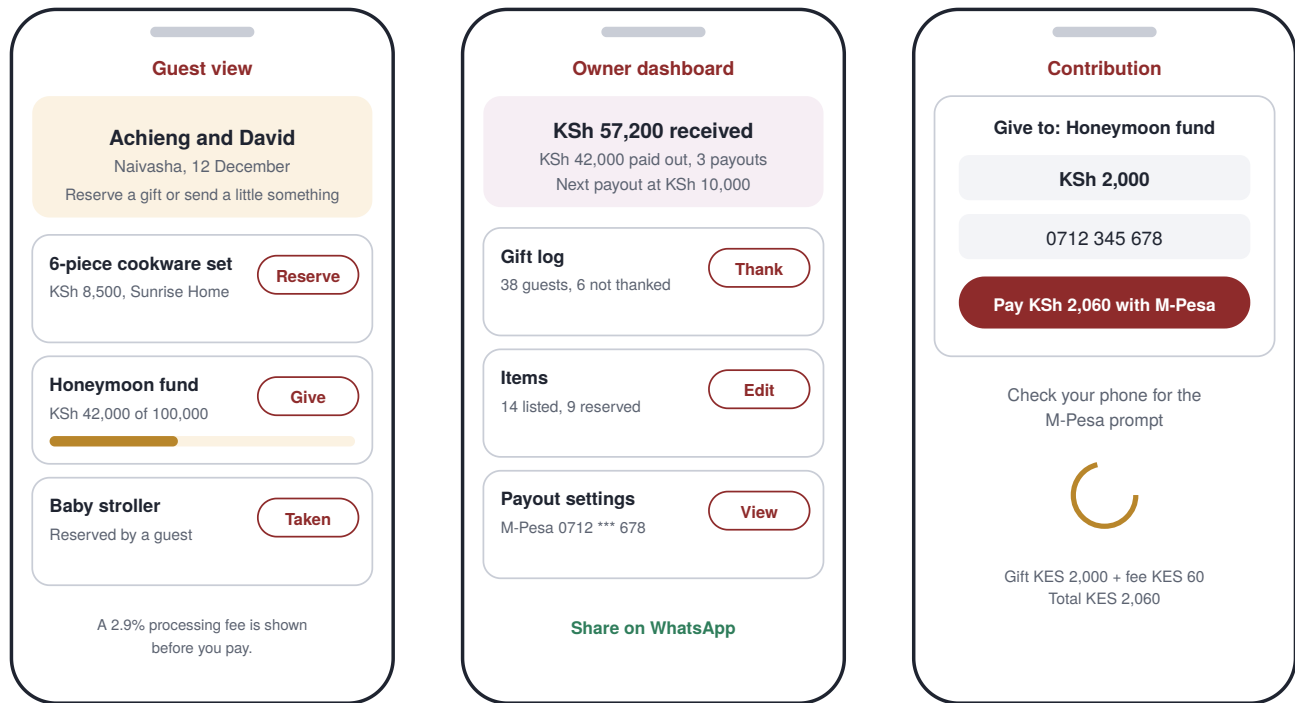


Figure 6. Key screens (low fidelity)

7.1 Hadia business onboarding



Figure 7. Hadia business onboarding

Step	Action	Responsible
1	Submit company documents: certificate of incorporation, CR12, KRA PIN, director identification, bank or paybill details.	Hadia
2	Create the Hadia tenant and assign the platform plan with escrow and M-Pesa features.	Codevertex
3	Set up Hadia's PayHero Team, complete KYC to tier 3 and save the Team's API key in treasury.	Hadia, PayHero
4	Synchronise the paybill or till and map the wallet and channels to the chart of accounts.	Codevertex
5	Accept the escrow terms in treasury.	Hadia
6	Configure commission, payout policy, approval limit, contribution limits and sweep schedule.	Hadia finance

7	Execute live KES 1 tests for contribution, payout, refund and sweep, and reconcile each.	Codevertex, Hadia
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7.2 Registry owner onboarding

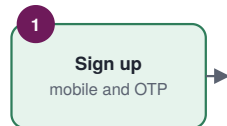


Figure 8. Registry owner onboarding

Step	Action	System behaviour
1	Owner enters a mobile number	One-time password sent by SMS, WhatsApp fallback. Codes hashed, 5-minute expiry, rate limited per number and IP. AUTO
2	Owner confirms age and accepts terms	Version and timestamp recorded
3	Owner creates the registry	Type, title, date, location, cover photo, description
4	Owner verifies the payout number	One-time password sent to the payout number
5	Registry published	Escrow account opened in treasury with default rules AUTO
6	Owner shares the registry	Short link and preview image generated AUTO

Exception: if the escrow account cannot be opened, the registry is saved as a draft and the request is retried in the background. Item entry continues uninterrupted.

7.3 Item listing

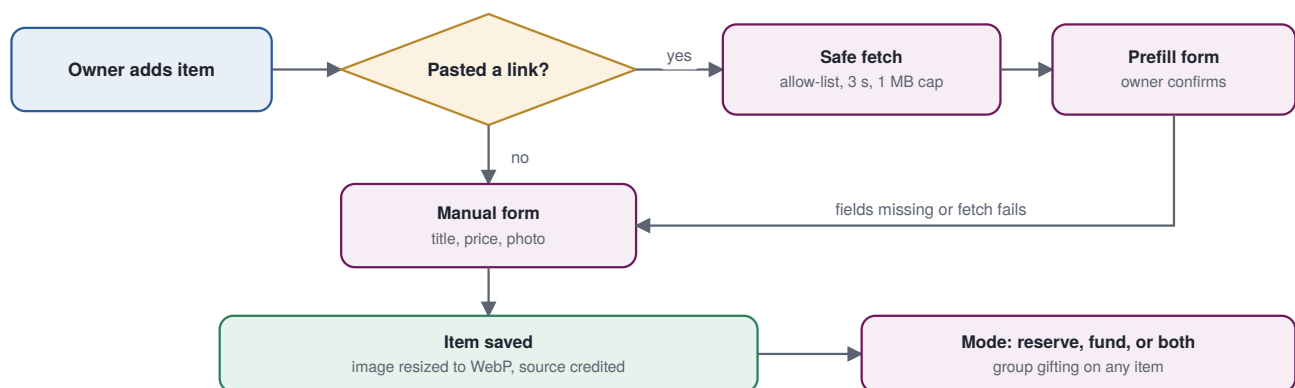


Figure 9. Item listing

1. The owner pastes a product link or selects manual entry.
2. For links, the server retrieves public page metadata under strict controls: public addresses only, 3-second timeout, 1 MB limit, no cookies.
3. Title, image and price are prefilled for owner confirmation. Where no data is found, the manual form is shown.
4. Photos are converted to WebP at 640 and 1280 pixels, stored in object storage and served through the CDN. The source store is credited.
5. The owner sets the item mode: reserve, fund, or both.

7.4 Sharing

1. Each registry receives a short link; premium registries may set a custom link.
2. A 1200 by 630 pixel preview image under 300 KB is generated for WhatsApp and social platforms. **AUTO**
3. Sharing options: WhatsApp, SMS, copy link and QR code.

7.5 Item reservation

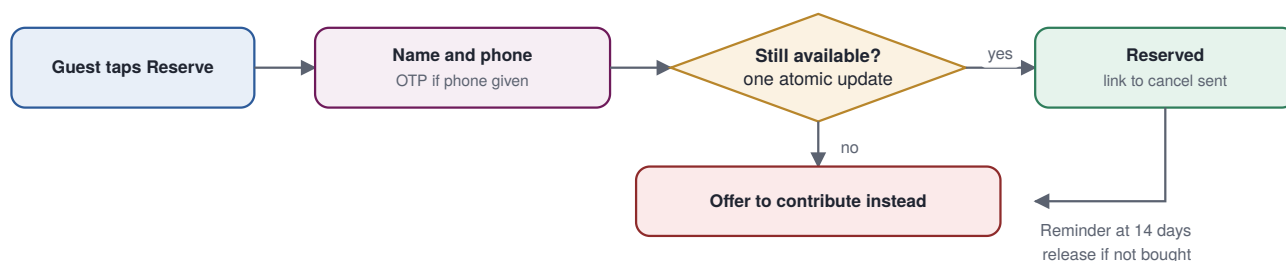


Figure 10. Item reservation

1. The guest selects Reserve and enters a name and, optionally, a mobile number.
2. The reservation is recorded in a single conditional update. Concurrent requests for the last unit result in one reservation; other guests are offered a contribution instead.
3. The guest receives a cancellation link. The item is shown as reserved to all guests.
4. After 14 days without purchase confirmation, reminders are sent and the owner may release the item. **AUTO**

7.6 Cash contribution

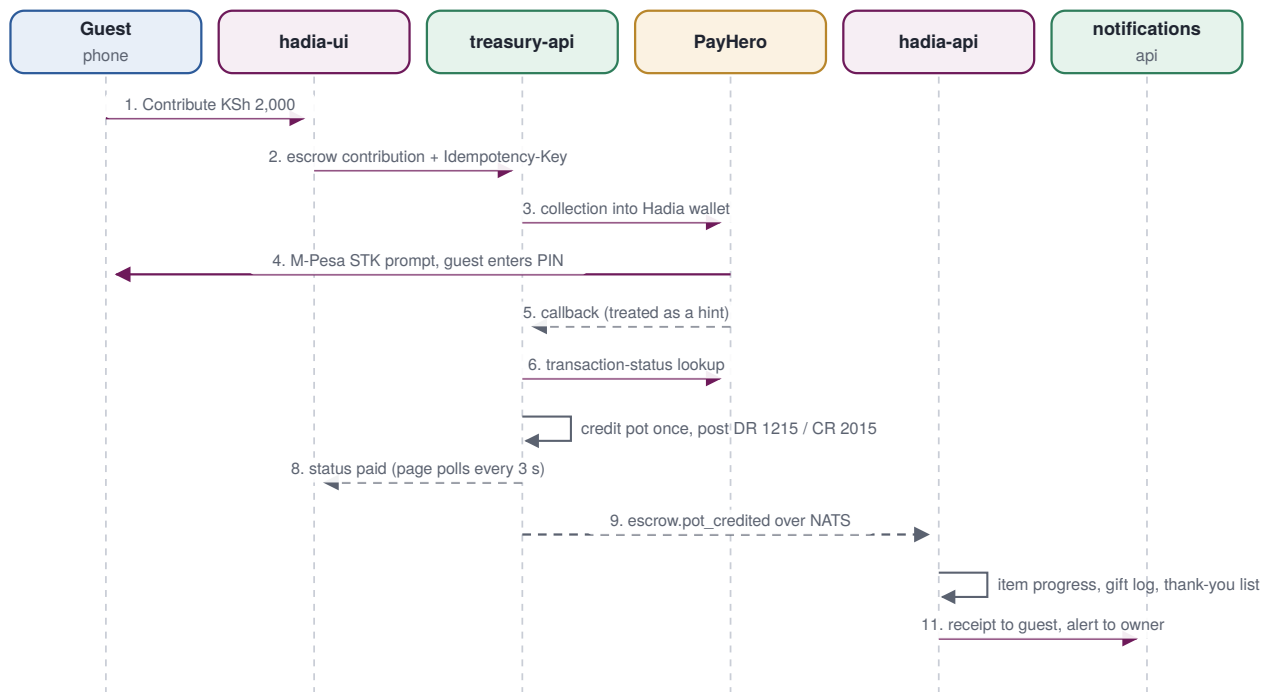


Figure 11. Cash contribution sequence

1. The guest selects an item or fund and enters an amount and mobile number. Limits are validated and the total, including any processing fee, is shown.
2. hadia-ui submits the contribution to treasury with an idempotency key.
3. Treasury initiates a PayHero collection into the Hadia wallet. The guest authorises the STK push.
4. On callback, treasury verifies the transaction with PayHero, credits the registry once and posts the ledger entry. **AUTO**
5. The checkout page confirms payment. Treasury publishes the credit event.
6. hadia-api updates item progress and the gift log and triggers the guest receipt and owner notification. **AUTO**

Exceptions: failed or cancelled prompts leave the contribution pending and the paybill option is offered. Repeated failures to one number (five in 24 hours) suspend further prompts to that number. Missing callbacks are resolved by status enquiry.

7.7 Registry management

1. The dashboard shows amounts received, paid out, pending and the next payout.
2. Owners may change the target, threshold (within Hadia's limits) and event date. The payout number changes only after verification of the new number and never during an active payout.
3. Contributions above an item's price remain in the registry balance and are paid out with it.
4. After the hold period the registry closes to new contributions. Owners may reopen it.

7.8 Registry lifecycle rules

Question	Rule
Purchase confirmation	The reservation message contains a link to mark the gift as purchased, with an optional receipt photo. The owner sees "Reserved" or "Purchased" and can confirm receipt. Unconfirmed reservations trigger reminders after 14 days.
Reserved items view	The owner dashboard filters items by Available, Reserved, Purchased and Funded, with the reserving guest's name and date.
Manual release	The owner can release any reservation. The guest is notified and the item becomes available again.
Registry transfer	Ownership can be transferred to another person. The new owner verifies their mobile number and payout number by one-time password. Transfers are blocked while a payout is in progress and are recorded in the audit log.
Multiple beneficiaries	Each registry pays out to one verified M-Pesa number. Co-owners can manage the registry but payouts go to the designated beneficiary. Split payouts across several beneficiaries can be added in a later phase.
Event date change	Allowed at any time, including after contributions begin. The payout hold date moves with the event date, within the 180-day maximum hold. Guests are not charged again and existing contributions are unaffected.
Registry expiry	30 days after the event date the registry closes to new contributions and reservations. Any remaining balance is paid out automatically. The registry stays visible to the owner in read-only mode.
Reopening	A closed registry can be reopened by the owner within 90 days, or by Hadia at any time. Cancelled registries cannot be reopened.

7.9 Automated payout

Registries meeting a payout rule are paid every 5 minutes to the verified number, net of commission, with approval required only at or above Hadia's limit. The owner receives an SMS showing gross amount, commission and net amount. **AUTO** Full process in [Funds management and payouts](#).

7.10 Commission and sweep

Commission is deducted at payout, recorded as Hadia income and transferred daily to Hadia's business account above the sweep minimum. **AUTO** eTIMS invoices for Hadia's fees are issued through treasury's KRA integration.

7.11 Premium upgrade

1. The owner selects Upgrade and is shown the features and price.
2. hadia-api creates a payment intent in treasury and opens the shared payment modal.
3. On confirmation, treasury records the sale as Hadia revenue and hadia-api enables the custom link, printable list and instant payouts. **AUTO**
4. A receipt is issued. Refunds follow the refund policy.

7.12 Cancellation and refunds

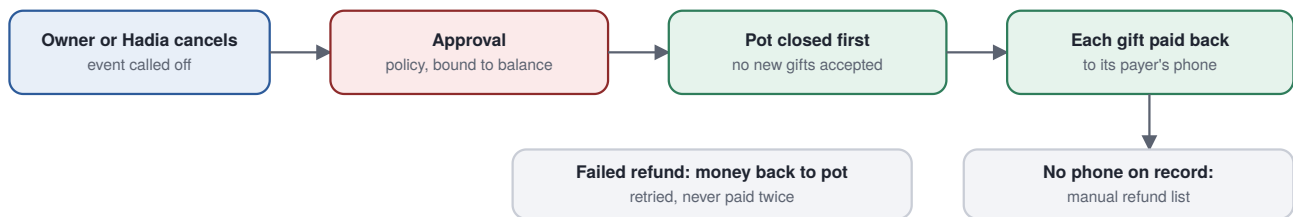


Figure 12. Cancellation and refunds

1. The owner requests cancellation, or Hadia cancels following a fraud or policy finding.
2. The refund is authorised under the approval policy, bound to the registry and its balance.
3. The registry is closed to new contributions. Each contribution is refunded to the paying number under its own reference. **AUTO**
4. Failed refunds return to the registry balance for retry. Repeated cancellation requests never duplicate a refund. Contributions without a number are listed for manual settlement.

7.13 Exceptions and reconciliation

1. Reconciliation runs hourly. **AUTO**
2. The exception queue lists failed payouts, name mismatches, manual refunds, reconciliation variances, reported registries and pending approvals.
3. Each entry records the facts, recommended action and resolver. Finance receives a daily summary.

7.14 Thank-you tracking

1. All contributions and reservations appear in the gift log with guest messages.
2. The owner sends a prepared WhatsApp thank-you and marks the guest as thanked.
3. A reminder listing outstanding thank-yous is sent one week after the event. Owners may disable it.

7.15 Data subject requests

1. An owner or guest submits a request for access, correction or deletion.
2. Identity is confirmed by one-time password and the statutory deadline is recorded.
3. Access requests produce a machine-readable export.
4. Deletion erases or anonymises personal data. Financial records required by law are retained for the statutory period under a reference, and the requester is informed.

SECTION 8

Data design

hadia-api owns the registry domain and references other services by ID: the owner's user ID in auth-api and the escrow account and payment intent IDs in treasury.

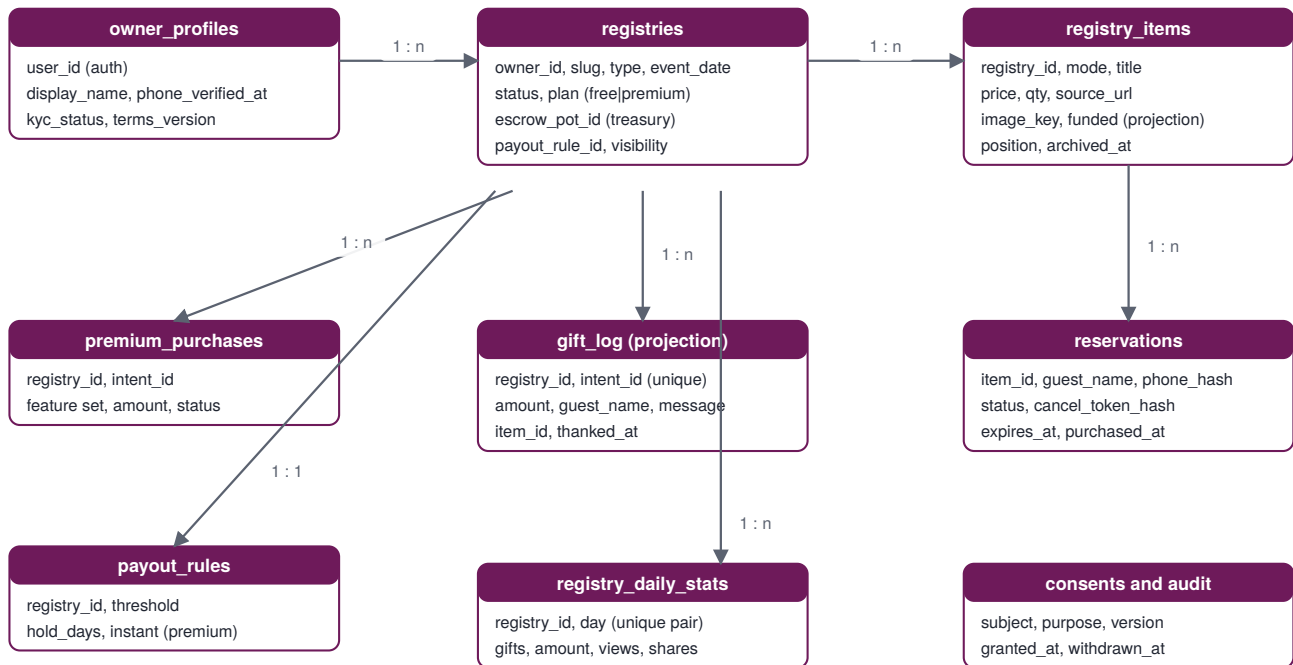


Figure 13. hadia-api data model

8.1 Scalability measures

Measure	Purpose
Tenant ID and query-specific indexes on every table, managed by versioned Atlas migrations	Consistent query performance and controlled schema change
Keyspace pagination on all lists	Constant response time regardless of page depth
Unique constraint on payment intent ID in the gift log	No duplicate records from redelivered events
Counters updated in a single SQL statement	Correct totals under concurrency
Daily aggregate table for reporting	Reports independent of raw data volume
Guest numbers hashed for matching and encrypted where display is required	Data protection
Images in object storage; keys only in the database	Small, fast database

SECTION 9

Interfaces, events and notifications

9.1 hadia-api endpoints

Area	Endpoints	Access
Registries	GET, POST /registries; GET, PATCH /registries/{id}; POST /registries/{id}/publish, /close, /cancel	Owner
Items	GET, POST /registries/{id}/items; PATCH, DELETE /items/{id}; POST /items/preview; POST /items/reorder	Owner
Public	GET /r/{code}; POST /r/{code}/items/{item}/reserve; POST /reservations/{token}/cancel	Public, rate limited
Gifts	GET /registries/{id}/gifts; POST /gifts/{id}/thanked; GET /registries/{id}/statement	Owner
Premium	POST /registries/{id}/premium	Owner
Privacy	POST /privacy/export, /privacy/delete; GET /privacy/requests	Owner; guest by OTP
Administration	Search, moderation, exceptions, settings, reports	Staff by role

9.2 Treasury endpoints

Function	Endpoint
Escrow account create, read, update	POST, GET, PUT /{tenant}/escrow/pots
Guest contribution	POST /api/v1/pay/{tenant}/escrow/{code}/contributions
Public account view	GET /api/v1/pay/{tenant}/escrow/{code}
Payout; cancellation with refunds	POST /{tenant}/escrow/pots/{pot}/release; /cancel?refund=true
Commission rule, readiness, summary	GET, PUT /{tenant}/escrow/fee-rule; GET /escrow/readiness; GET /escrow/summary
Statements	GET /{tenant}/escrow/pots/{pot}/statement; /statement/document
Premium sale	POST /{tenant}/payments/intents

9.3 Events

Event	Publisher	hadia-api action
escrow.pot_created	treasury	Mark escrow ready; publish registry
escrow.pot_credited	treasury	Update gift log, item funding and statistics; send receipt and notification
escrow.dispatched	treasury	Record payout; notify owner of gross, commission and net
escrow.dispatch_failed	treasury	Raise exception; notify owner of retry
escrow.cancelled	treasury	Close registry; notify guests of refunds

payment.succeeded	treasury	Activate premium features
hadia.registry_published, hadia.reservation_created	hadia-api	Consumed by notifications and analytics

Consumers are idempotent on event ID. Events are retained in the outbox during broker outages.

9.4 Notifications

Message	Recipient	Channel
One-time password	Owner, guest	SMS; WhatsApp fallback
Contribution receipt	Guest	SMS or WhatsApp
New contribution or reservation	Owner	WhatsApp or SMS; daily digest option
Payout completed or delayed	Owner	SMS
Reservation and thank-you reminders	Guest, owner	WhatsApp or SMS
Refund completed	Guest	SMS
Exception summary, approval request	Hadia finance	Email, WhatsApp
Product updates	Opted-in owners	Email or WhatsApp with unsubscribe

SECTION 10

Reporting and analytics

Report	Audience	Source
Registry dashboard	Owner	Treasury balances and payouts; gift log
Contributions and value by period	Hadia	Daily aggregates
Active registries, new registries, conversion	Hadia	Daily aggregates and page view counters
Commission and premium revenue	Hadia finance	Treasury ledger, trial balance, income statement
Funds held, in progress and available	Hadia finance	Treasury escrow summary
Payout performance	Hadia operations	Payout records: time to completion, failure reasons
Registry statement	Owner, finance	Treasury statement in PDF, CSV, Excel

Analytical queries run on aggregate tables and may use the read replica. Dashboard results are cached for 60 seconds and invalidated by the related event.

SECTION 11

Security

Threat	Controls
Account takeover	OTP rate limits by number and IP, hashed codes, short expiry, session revocation, MFA for staff, re-verification for payout changes
Payout redirection	Payout number verified by OTP, changes blocked during active payouts, approvals bound to amount and destination, name check on first payout, change alerts
Duplicate charges or payouts	Idempotency keys, unique payout references, single settlement path, provider verification of callbacks
Fraudulent registries	Moderation, guest reporting, registry creation limits, balance caps, first-payout hold for new owners
Money laundering	Contribution and balance limits, KYC, verified payout numbers, pattern monitoring, escalation to Hadia's compliance contact
Server-side request forgery	Public address allow-list with post-resolution DNS checks, redirect validation, time and size limits
Injection and cross-site scripting	Parameterised queries, output encoding, Content Security Policy, sanitised input
Automated abuse	Rate limiting, STK prompt guard, Cloudflare WAF
Data exposure	TLS, encryption at rest, least-privilege roles, payout details excluded from public pages, staff access logging
Card data	Provider-hosted card checkout; no card data on Hadia systems

11.1 Production access

Control	Implementation
Who has access	A named list of Codevertex platform engineers, provided to Hadia and updated within 5 working days of any change. No shared accounts.
Server access	Key-based SSH only, restricted by host firewall (default deny; only SSH, HTTP and HTTPS open)
Changes to production	Only through version-controlled GitOps deployment (ArgoCD); manual changes are reverted automatically. Every change is attributable to a reviewed commit.
Secrets	Encrypted as sealed secrets; separate database credentials per service; payment provider keys encrypted in the database
Network isolation	Kubernetes network policies on payment, authentication and notification services; pod security standards enforced
Hadia staff	Role-based access through single sign-on with MFA; finance approvals require a one-time password
Audit	Staff logins, approvals, settings changes and financial actions recorded in tamper-evident audit logs, available to Hadia

11.2 Codevertex access to Hadia data

Codevertex staff do not access Hadia's personal or financial records in normal operation. Access occurs only to resolve a support request raised by Hadia, to investigate an incident, or to perform a restore, and is limited to the records concerned. Each such access is logged and reported to Hadia on request. These obligations are set out in the data processing agreement, and every engineer with production access is bound by a confidentiality undertaking.

11.3 Security incident notification

Event	Timeline
Codevertex notifies Hadia of a confirmed personal data breach	Within 24 hours of confirmation
Hadia notifies the Data Protection Commissioner (section 43, Data Protection Act)	Within 72 hours of becoming aware; Codevertex supplies the facts required
Affected users notified where there is a real risk of harm	Without undue delay, with Hadia's approval of the message
Written incident report: cause, impact, remediation	Within 5 working days

11.4 Penetration testing

An internal security review against the OWASP Application Security Verification Standard is included in week 11. An independent penetration test by an accredited third party is recommended before launch and annually thereafter. It is commissioned at Hadia's cost at the tester's rate; Codevertex supports the test and remediates all high and critical findings at no charge before launch.

11.5 Data processors

Processor	Purpose	Data received	Location
Codevertex Africa Limited	Software, hosting operation, support	All platform data	Kenya (operations); EU (servers)
Contabo GmbH	Server infrastructure	Platform data stored on its servers	Germany (EU)
Cloudflare, Inc.	Edge network, firewall, offsite backups	Traffic in transit; backup files	Global edge; EU and US
PayHero	Payment collection, wallet, payouts	Payer number, amount, payout details	Kenya
Safaricom PLC	M-Pesa	Payer and recipient number, amount	Kenya
Africa's Talking	SMS delivery	Recipient number, message text	Kenya
Meta Platforms	WhatsApp messaging	Recipient number, message text	Global

SECTION 12

Hosting, backup and disaster recovery

This section states where Hadia's data is held, how it is backed up and how service is restored after a failure.

12.1 Hosting location

Component	Provider and location	Content
Application and database servers	Contabo Cloud VPS, European Union region (Germany). Dedicated virtual server: 12 vCPU, 47 GB RAM, 484 GB NVMe storage, running Kubernetes (k3s)	hadia-ui, hadia-api, treasury, authentication, notifications; PostgreSQL databases
Offsite backups	Cloudflare R2 object storage, separate provider and infrastructure; encrypted at rest	Nightly encrypted database backups
Edge network	Cloudflare (TLS termination, CDN, web application firewall)	Traffic in transit and cached public assets; no database content
Email	Codevertex mail server (mx1.codevertexafrica.com) on the same platform	Outbound email; no third-party email processor
Funds	PayHero, Kenya	Gift funds and payment records. Funds never reside on Codevertex infrastructure.

Hosting outside Kenya is a cross-border transfer under sections 48 to 50 of the Data Protection Act. The transfer is documented in the DPIA with its safeguards: provider security certification, encryption in transit and at rest for backups, access restriction and a data processing agreement. Hosting in a Kenyan data centre is available on request and would be quoted as a change to the hosting cost.

12.2 Backup schedule

Backup	Frequency	Scope	Storage	Retention
Platform recovery backup	Daily, 05:00 EAT	All platform databases, verified for integrity after each run	Server disk and Cloudflare R2 (offsite)	Each new copy replaces the previous one only after it is verified and confirmed offsite
Hadia tenant backup	Daily, 05:00 EAT	Hadia's registry and treasury data only, encrypted	Cloudflare R2 (offsite), with optional mirror to Hadia's own storage (Google Drive, OneDrive, S3, SFTP)	7 daily and 4 weekly copies
Payment records	Continuous	Every collection and payout	PayHero and Safaricom, independently of Codevertex	Provider retention

12.3 Recovery objectives

Objective	Target	Meaning for Hadia
Recovery point objective (RPO)	24 hours	In the worst case, total loss of the server, changes made since the last nightly backup are restored from other sources. Contributions and payouts in that window are recovered from PayHero's transaction records and re-reconciled, so no payment record is lost. Registry edits, item changes and reservations made in that window may need to be re-entered.
Recovery time objective (RTO)	4 hours	Service is restored on a replacement server within 4 hours of a declared disaster.
Optional enhanced RPO	15 minutes	Continuous database log archiving to offsite storage, quoted on request.

12.4 Restore testing

A restore drill is performed quarterly: the latest offsite backup is restored into an isolated environment, record counts and ledger balances are verified against production, and a short report is shared with Hadia. The first drill takes place before launch, in week 11.

12.5 Platform outage response

Scenario	Impact	Response
Service or pod failure	Seconds to minutes	Automatic restart; two replicas per service; health checks every 2 minutes with alerting
Server failure or provider outage	Platform unavailable	Automated rebuild on a replacement server from version-controlled configuration and backed-up encryption keys, followed by restore of the latest backup and reconciliation with PayHero.
During any outage	Guests cannot contribute; payouts pause	Funds remain safe in Hadia's PayHero wallet. Pending payouts resume automatically on recovery. Hadia is notified within 30 minutes of a confirmed incident and updated every 2 hours.

Capacity scales by adding nodes to the cluster under a documented procedure, without service interruption. Additional capacity is planned with Hadia as monthly contributions grow.

SECTION 13

Legal and regulatory compliance

This chapter sets out the legal obligations applicable to Hadia and the controls that address them. It does not constitute legal advice. Items marked **Confirm** require an opinion from Hadia's advocate before launch.

13.1 Roles and responsibilities

Party	Role	Obligations
Hadia	Service operator and data controller; holds customer funds for beneficiaries through its PayHero account	User terms, fees, payouts, complaints, ODPC registration, tax, KYC and AML cooperation
Codevertex Africa Limited	Software provider and data processor	Secure development and operation, processing on Hadia's instructions under a data processing agreement, breach notification to Hadia
PayHero	Payment service provider	Collections, wallet custody, payouts, KYC and AML, card processing
Africa's Talking, Meta (WhatsApp), SMTP email host	Sub-processors	Message delivery under their terms

13.2 Applicable law

Law	Obligation	Control
Constitution of Kenya, Article 31; Data Protection Act, 2019 and Regulations, 2021	Lawful and minimal processing, ODPC registration, DPIA, data subject rights, 72-hour breach notification, transfer safeguards, consent for direct marketing, parental consent for children's data	Privacy notice, consent records, data minimisation, DPIA, ODPC registration, data request workflow, breach procedure, documented hosting location, opt-in marketing
Consumer Protection Act, 2012	No misleading representations; for internet agreements, disclosure before acceptance, express acceptance, error correction and a copy of the agreement	Fees and payout timing disclosed before sign-up and payment, versioned click-through acceptance, downloadable terms, verified marketing claims
Competition Act, 2010	No false or misleading representations, including reviews and testimonials	Consented testimonials only; claims supported by records
National Payment System Act, 2011 and Regulations	Payment services and holding funds for others are regulated by the Central Bank of Kenya	Funds held and paid by PayHero; capped hold periods and balances; per-registry ledgers; hourly reconciliation. Confirm whether Hadia requires CBK authorisation or guidance.
Proceeds of Crime and Anti-Money Laundering Act, 2009	Prevent laundering; maintain records; cooperate with reporting institutions	KYC tier 3, verified payout numbers, contribution and balance caps, transaction monitoring, record retention, designated compliance contact

Computer Misuse and Cybercrimes Act, 2018	System and data protection; handling of unlawful content	Security controls, audit logging, moderation and reporting, incident response
Kenya Information and Communications Act; Consumer Protection Regulations, 2010	Validity of electronic records; no unsolicited messages; opt-out	Logged acceptance, transactional messaging by default, STOP and unsubscribe on marketing, registered sender ID
Tax Procedures Act, 2015; VAT Act, 2013; eTIMS	Record retention of at least five years; electronic tax invoices; VAT registration above threshold	Ledger in treasury, gifts recorded as liabilities, commission and premium as income, eTIMS invoicing. Confirm VAT and excise treatment of fees.
Children Act, 2022; Data Protection Act, section 33	Parental consent and best interests for children's data	Owners aged 18 or over, parental confirmation before child names or photos, unlisted registries by default
Persons with Disabilities Act; Constitution, Article 54	Accessible public services	WCAG 2.2 AA
Copyright Act, 2001; Trade Marks Act	Respect for third-party content and marks	Licensed fonts and images, credited product thumbnails with takedown, descriptive use of store names only
Business Registration Service Act; Companies Act, 2015	Trade under registered name with business identification	Registered name, number, address and contacts on every page and receipt
EU and UK GDPR; US CAN-SPAM and COPPA (diaspora users)	Equivalent privacy rights, email unsubscribe, no data from children under 13	Kenyan controls applied, prior consent for optional cookies, unsubscribe in every email, service not directed at children

13.3 Compliance checklist

#	Requirement	Implementation	Owner
1	Privacy policy	Controller, processors, purposes, lawful bases, retention, rights, transfers and contacts. Linked from every page, form and message.	Codevertex draft; advocate approval
2	Terms of service	Owner, guest and escrow terms; acceptable use; fees; liability; disputes; Kenyan law. Versioned acceptance; 14 days' notice of changes.	Codevertex draft; advocate approval
3	Refund policy	Refund grounds, timelines, premium refunds within 14 days if unused, fee treatment. Enforced by the refund workflow.	Hadia
4	Cookie policy	All cookies listed by category, purpose and duration. Strictly necessary cookies only at launch.	Codevertex
5	Cookie consent	No pre-selected options; equal reject and accept controls; settings available from the footer.	Codevertex
6	Form consents	Purpose stated for each field; separate unticked consents; consent versioned and withdrawable.	Codevertex
7	Data minimisation	Guests provide name and number only. No identity numbers, dates of birth or location tracking.	Codevertex

8	Third-party audit	No advertising trackers. Register of third parties and data shared. Analytics only with consent.	Codevertex
9	No dark patterns	No pre-ticked boxes, false urgency or confirm-shaming. Cancellation and deletion as simple as sign-up.	Codevertex
10	No hidden fees	Guests see the processing fee and total before paying. Owners see commission and payout terms before creating a registry and on every payout.	Codevertex
11	Genuine reviews	Testimonials published only with written consent, retained on file.	Hadia
12	Substantiated claims	Marketing statements reflect actual funds handling and payout terms.	Hadia
13	Alternative text	Required for uploaded photos; descriptive text on all content images.	Codevertex
14	Colour contrast	4.5:1 for text and 3:1 for controls.	Codevertex
15	Keyboard navigation	Full keyboard operation, visible focus, managed dialogs, skip link, automated accessibility testing.	Codevertex
16	Business details	Registered name, number, address and contacts in the footer and on receipts.	Codevertex
17	Children's data consent	Owner age confirmation and parental consent before child information.	Codevertex
18	Unsubscribe	One-click unsubscribe in email; STOP in SMS and WhatsApp marketing.	Codevertex
19	Licensed fonts and images	Open-licence fonts and licensed imagery, recorded in a licence register.	Codevertex
20	Data deletion	Self-service export and deletion with deadline tracking and statutory retention.	Codevertex

13.4 Website content review

Current text	Issue	Recommendation
"Loved by 400+ families"	Unverified figure	Remove until supported by platform data
Achieng and David testimonial	Not a verified customer statement	Replace with consented pilot testimonials
"Funds go straight to you"	Funds are held and paid out by rule, net of commission	"Cash gifts are held securely and paid to your M-Pesa automatically, less Hadia's 4% fee."
"Your money, kept safe"	Requires substantiation	Link to a page describing custody, ledgers and reconciliation
"You only ever share in what's actually given"	Incomplete fee disclosure	"Free to create. 4% of cash gifts. Optional premium at KES 300 per event."
"Carrefour, Naivas, TikTok Shop"	Third-party marks	Text references only; no logos or implied partnership

13.5 Record retention

Record	Retention	Basis
Contributions, payouts, refunds, invoices, ledger	7 years	Tax Procedures Act; AML record keeping
Owner accounts and registries	Until deletion, or 24 months after last activity, then anonymised	Data minimisation
Reservation contact details	12 months after the event	Reservation and thank-you management
One-time passwords	5 minutes, hashed	Security
Consent and terms acceptance	Account life plus 5 years	Evidence of consent
Security and audit logs	12 months	Incident investigation

13.6 Legal deliverables

- Privacy notice, cookie policy, terms of service, refund policy, acceptable use policy
- Data processing agreement and sub-processor list
- Data protection impact assessment and record of processing activities
- Breach response procedure and data request procedure
- Third-party and licence registers, accessibility statement

SECTION 14

Project delivery

Delivery follows Agile Scrum with weekly demonstrations on the staging environment.

14.1 Methodology

Practice	Approach
Iterations	One-week Sprint 0, four two-week sprints, one-week Sprint 5, hardening week, launch week
Demonstrations	Weekly 30-minute review with the product owner; decisions logged in the backlog
Backlog	User stories traced to FR and NFR identifiers, with acceptance criteria
Definition of done	Code reviewed, tests and accessibility checks passed, documented, deployed to staging, accepted
Environments	Development, staging and production on the Codevertex platform
Quality assurance	Unit, integration, end-to-end and contract tests; load testing before launch
Release management	CI on GitHub Actions; GitOps deployment through ArgoCD
Reporting	Weekly written status: progress, next steps, risks, decisions required
Change control	Equal-size scope substitutions within the fixed price; other changes estimated and approved in writing

14.2 Project team

Role	Allocation
Project lead and business analyst	Part time, weeks 1 to 12
Full-stack engineers (Go, Next.js)	Two, weeks 1 to 12
Product designer	Part time, weeks 1 to 5; reviews thereafter
QA engineer	Part time, weeks 3 to 12
Platform and DevOps engineer	As required

14.3 Implementation schedule

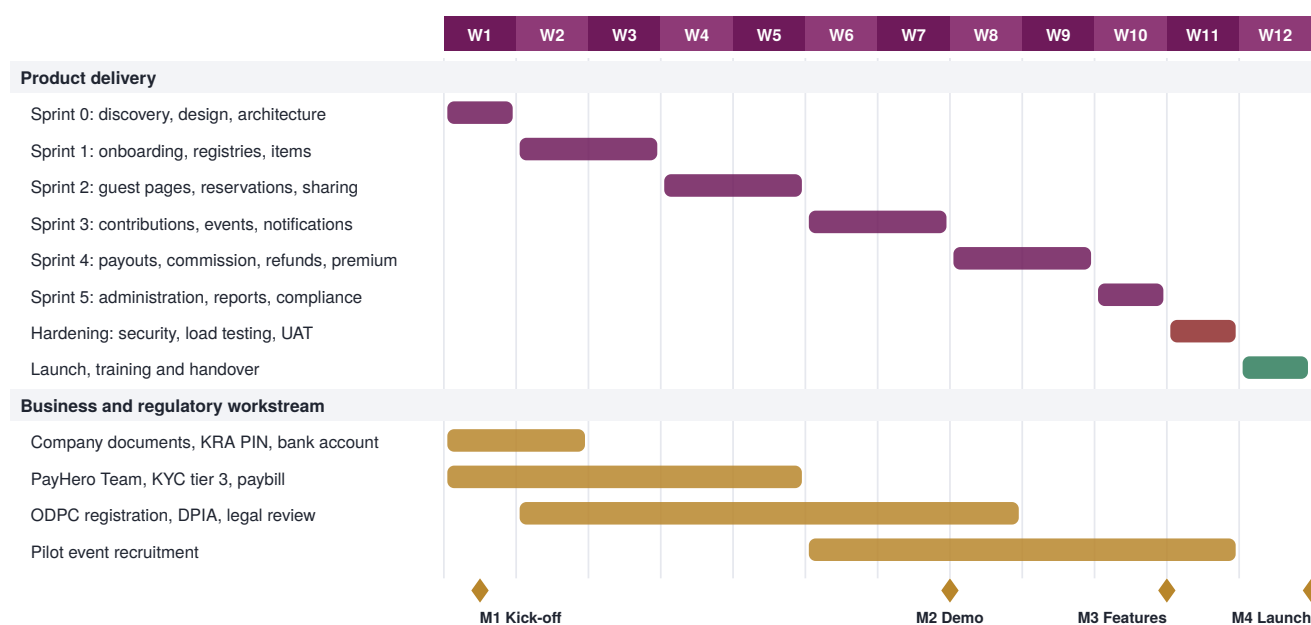


Figure 14. Implementation schedule

Week	Phase	Deliverables
1	Sprint 0	Kick-off, backlog, UX flows, visual design, architecture approval, environments, legal pack outline
2 to 3	Sprint 1	Mobile sign-in, owner onboarding, registries, manual items, image pipeline, dashboard, tenant configuration
4 to 5	Sprint 2	Guest pages, reservations, link completion, sharing and previews, escrow account creation, payout number verification
6 to 7	Sprint 3	Contributions, event processing, gift log, receipts and notifications, statements. Milestone 2: end-to-end demonstration with a live contribution
8 to 9	Sprint 4	Automated payouts, commission, sweep, contribution limits, approvals, cancellation and refunds, premium
10	Sprint 5	Administration console, exception queue, reports, consent and data requests, accessibility. Milestone 3: feature complete
11	Hardening	Security review, load test, UAT with pilot events, legal pack approval
12	Launch	Production release, training, handover. Milestone 4: launch acceptance

Based on contract signature by 16 October 2026, work begins on 19 October 2026 and launch is scheduled for the week of 4 January 2027. Public holidays are accounted for.

SECTION 15

Testing and acceptance

Test	Objective	Exit criteria
Automated tests	Business rules and payment logic	All tests pass on every merge
End-to-end tests	Owner, guest, refund and premium journeys	Pass on Android Chrome, iOS Safari and desktop
Live payment tests	Contribution, payout, refund and sweep	KES 1 transactions reconcile with PayHero and the ledger
Load test	Peak traffic	2,000 contributions per hour and 50 page views per second within NFR-02
Security review	OWASP risks, authentication, payout changes, link retrieval	No open high or critical findings
Accessibility	WCAG 2.2 AA	Automated and manual checks passed
User acceptance	Business acceptance by Hadia and pilot users	All M requirements accepted; no open severity 1 or 2 defects

SECTION 16

Commercial terms

16.1 Fixed price

Work package	KES
Discovery, UX and visual design, architecture	20,000
Registry service: registries, items, reservations, link completion, gift log, statistics	55,000
Web application: website, guest pages, owner dashboard, administration console	55,000
Funds integration: escrow, contributions, automated payouts, commission sweep, limits, refunds, premium	45,000
Mobile sign-in, message templates and preferences	15,000
Compliance features, legal pack drafts, accessibility	20,000
Testing, security review, UAT, launch and handover	30,000
Total (inclusive of taxes)	240,000

16.2 Payment schedule

Milestone	Due	Share	KES
M1 Contract signature	Week 0	30%	72,000
M2 End-to-end demonstration	End of week 7	30%	72,000
M3 Feature complete	End of week 10	30%	72,000
M4 Launch acceptance	End of week 12	10%	24,000
Total		100%	240,000

16.3 Monthly service fee

Service	KES per month
Treasury platform subscription: ledger, escrow, payouts, approvals, reconciliation, eTIMS invoicing	Included
Platform services: authentication, notifications engine, events and job scheduling	Included
Monitoring, security updates and incident response	Included
Dedicated support and maintenance: support escalation, defect correction, configuration changes	Included
Total monthly service fee (inclusive of taxes)	8,000

The monthly service fee is billed from the launch month.

Term	Position
Usage limits	No per-transaction charge from Codevertex. The fee covers up to 20,000 contributions and 10,000 active registries per month under fair use.

Volume growth	Above the fair-use level, server capacity is reviewed with Hadia. Any additional capacity is charged as hosting, at cost, with 60 days' written notice and Hadia's agreement.
Price changes	Fixed for the first 12 months from launch. Thereafter it may be adjusted once a year by no more than 10%, with 60 days' written notice.

16.4 Client-borne costs

The following costs are separate from the build price and the monthly service fee and are paid by Hadia.

Item	Scope	Rate (KES)
Hosting	Application hosting, database, object storage, daily backups and CDN on the Codevertex platform, billed annually	12,000 per year
Domain	Registration and annual renewal of the Hadia domain	Registrar rate
SMS	One-time passwords, receipts, alerts and reminders; prepaid credit, minimum top-up KES 500	1.00 per SMS
WhatsApp	Receipts, alerts, thank-you messages and fallback one-time passwords; monthly plan	500 to 2,000 per month
Payment provider	PayHero payout fees, and processing fees if Hadia absorbs them	PayHero rates

16.5 Payment provider charges

Charge	Rate	Paid by
Gift contribution (into the wallet)	2.9% of the amount	Guest by default, or Hadia
Payout to an owner or to Hadia	PayHero payout band; KES 1.16 on a KES 20 payout	Hadia, from the wallet
Premium purchase (into Hadia's paybill or till)	PayHero band fee; KES 6 on KES 50, KES 0 up to KES 10	Guest by default; drawn from Hadia's service wallet
Card payments	PayHero card rate, confirmed at onboarding	As for gifts

Rates were measured on live PayHero transactions in October 2026. Treasury records the actual fee on every payment, so Hadia's reports show real costs. For premium purchases, treasury picks the cheaper of the two PayHero routes automatically, and Hadia tops up the service wallet from the treasury console.

16.6 Complete fee structure

Fee	Payer	Payee	Amount
Gift contribution	Guest	Registry (escrow)	Gift amount
Processing fee	Guest (default) or Hadia	PayHero	2.9% of the gift
Commission	Registry owner	Hadia	4% of gross (configurable)
Premium upgrade	Registry owner	Hadia	KES 300 per registry
Build price	Hadia	Codevertex	KES 240,000 one-time

Monthly service fee	Hadia	Codevertex	KES 8,000 per month
Hosting	Hadia	Codevertex	KES 12,000 per year
Domain	Hadia	Registrar	Registrar rate per year
SMS and WhatsApp	Hadia	Codevertex notifications	KES 1.00 per SMS; WhatsApp plan
Payout fees	Hadia	PayHero	PayHero payout band

16.7 Operating cost examples

Monthly item (KES)	1,000 gifts	5,000 gifts	10,000 gifts
Monthly service fee	8,000	8,000	8,000
Hosting (KES 12,000 per year)	1,000	1,000	1,000
SMS receipts (1 per gift)	1,000	5,000	10,000
WhatsApp Pro	2,000	2,000	2,000
Payout fees (1 payout per 10 gifts, about KES 25 each)	2,500	12,500	25,000
Operating cost, guests pay the processing fee	14,500	28,500	46,000
Commission at 4% (average gift KES 2,000)	80,000	400,000	800,000
Commission after costs	65,500	371,500	754,000
Commission after costs, if Hadia absorbs the 2.9% fee	7,500	81,500	174,000

Average gift KES 2,000. The payout fee per payout will be confirmed with PayHero for Hadia's account. Premium revenue is excluded.

16.8 Messaging rates

Plan or item	Allowance	KES
SMS	Per SMS of up to 160 characters	1.00
WhatsApp Basic	100 messages per month	500 per month
WhatsApp Standard	500 messages per month	1,000 per month
WhatsApp Pro	Unlimited messages	2,000 per month

Guest receipts are sent by SMS, one per contribution. Owner alerts, digests and thank-you messages are sent by WhatsApp. The WhatsApp Pro plan is recommended from launch because it carries no message cap, so WhatsApp cost stays fixed as volume grows.

16.9 Optional services

Service	Scope	Billing
Managed domain	Registration of the Hadia domain on Hadia's behalf, annual renewal, DNS management and SSL certificates	Annually, at the registrar's rate

Legal fees and marketing are excluded.

SECTION 17

Ownership, licensing and exit

Hadia owns its product, its data and its money. Codevertex retains the shared platform that Hadia uses under licence.

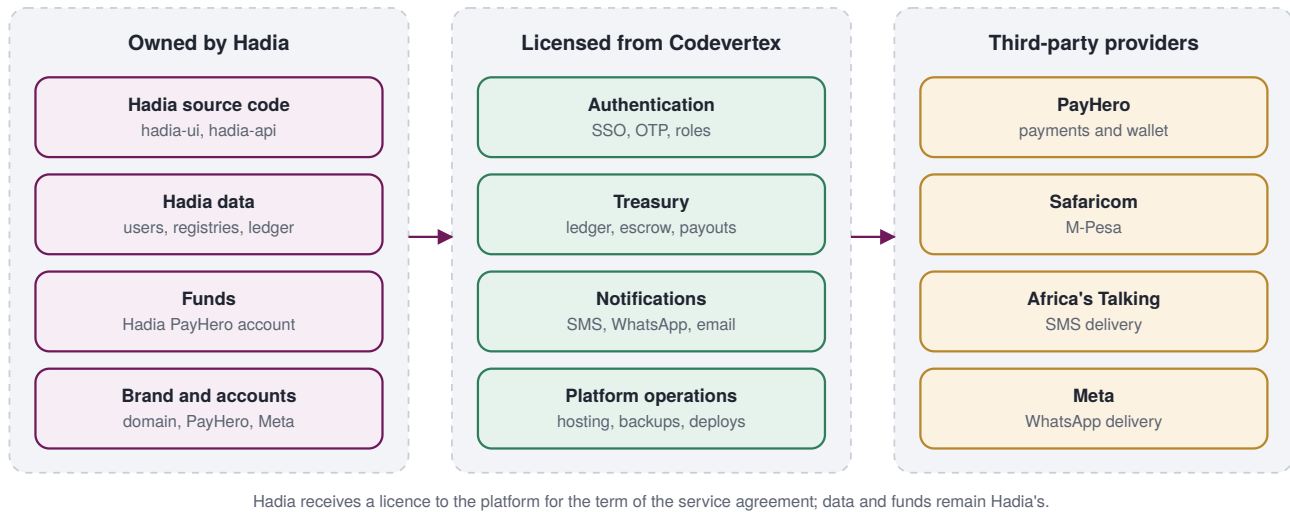


Figure 15. Ownership boundaries

17.1 Ownership

Asset	Owner	Terms
Hadia source code: hadia-ui, hadia-api, Hadia message templates, designs and content	Hadia	Intellectual property assigned to Hadia on payment of the build price in full
Hadia data: users, registries, items, reservations, contributions, payouts, ledger entries, consents	Hadia	Owned by Hadia from creation; Codevertex processes it only on Hadia's instructions
Gift funds and commission	Hadia (for registry owners)	Held in Hadia's PayHero account, never by Codevertex
Hadia brand, domain and accounts with PayHero, Safaricom and Meta	Hadia	Registered in Hadia's name
Codevertex platform: authentication, treasury (ledger, escrow, payouts), notifications, subscriptions, shared libraries, deployment tooling	Codevertex	Proprietary; licensed to Hadia as below

17.2 Repository access

The hadia-ui and hadia-api repositories are created in a GitHub organisation owned by Hadia, with Codevertex as a contributor. Hadia has read access from day one and full administrative control at handover. Codevertex platform repositories remain private to Codevertex.

17.3 Platform licence

Hadia receives a non-exclusive, non-transferable licence to use the Codevertex platform services for the duration of the service agreement, included in the monthly service fee. Hadia's own developers, or another vendor appointed by Hadia, may use the documented platform APIs with API keys issued to Hadia.

17.4 Appointing another developer

Hadia may appoint another developer at any time to maintain or extend hadia-ui and hadia-api. Codevertex provides API documentation, environment access for Hadia's appointed developer and a handover session. The platform services continue to be operated by Codevertex while the service agreement is in force.

17.5 Exit and migration

Step	Commitment
Notice	Codevertex gives at least 90 days' written notice before discontinuing any platform service Hadia relies on.
Data export	Complete export in open formats: PostgreSQL dump of the Hadia database, ledger and escrow statements (CSV and Excel), user and registry records (JSON), uploaded images
Funds	Unaffected. Funds sit in Hadia's PayHero account and continue to be accessible directly through PayHero.
Transition assistance	Up to 20 hours of engineering support to move to a replacement payments and ledger service, included; further support at the standard rate
Source code escrow (optional)	Platform source code required to run Hadia can be deposited with an independent escrow agent, released to Hadia if Codevertex ceases to trade. Agent fees are borne by Hadia.

17.6 Handover on termination

- Full ownership and administrative control of the hadia-ui and hadia-api repositories
- Final data export and the latest backups, delivered within 10 working days
- Architecture, API and operations documentation, and the legal pack
- Transfer of domain and DNS administration, where managed by Codevertex
- Written confirmation of deletion of Hadia data from Codevertex systems after export, subject to statutory retention

SECTION 18

Contract terms

Principal terms of the master services agreement between Hadia and Codevertex. Codevertex's standard terms apply where not varied below.

Term	Position
Initial term	12 months from launch, renewing monthly thereafter
Termination for convenience	Either party, after the initial term, with 60 days' written notice
Termination for cause	Either party, for a material breach not remedied within 30 days of written notice, or on insolvency
Monthly fee after termination	Payable to the end of the notice period only; no fees thereafter. Export and handover are provided at no charge.
Service level	99.5% monthly availability for guest pages, checkout and payouts, excluding scheduled maintenance notified 48 hours in advance
Service credits	Below 99.5%: 10% of the monthly fee. Below 99.0%: 25%. Below 95.0%: 50%.
Prolonged downtime	An outage exceeding 72 continuous hours, or service credits in any 2 consecutive months, entitles Hadia to terminate without notice period and receive full handover.
Delay caused by Codevertex	2% of the build price credited for each full week of delay to launch caused solely by Codevertex, up to 10% (KES 24,000). The final milestone is not due until launch acceptance.
Delay caused by Hadia	Applies where Hadia's decisions, documents, content, feedback, test participation or PayHero and KYC submissions are later than the agreed dates (decisions within 2 working days, documents within 3), or where Hadia changes scope. The schedule extends day for day and no delay credits apply. Milestone payments fall due on their original dates for work completed. Team standby time beyond a cumulative grace of 5 working days is charged at KES 20,000 per week, pro rata.
Force majeure and third parties	Events outside both parties' control: natural events, national network or power failures, regulatory action, and processing delays by PayHero, Safaricom or other providers after Hadia has submitted complete documents on time. The schedule extends day for day with no credits or charges to either party. If the event continues beyond 60 days, either party may terminate, and Hadia pays for work completed to that date.
Launch without payments	Where payment onboarding is the only outstanding item, the platform can launch with reservations; payments are enabled once PayHero is live, and launch acceptance is not withheld on that account.
Liability cap	Each party's total liability is limited to the fees paid by Hadia in the 12 months preceding the claim. Indirect and consequential losses are excluded.
Exclusions from the cap	Breach of confidentiality, the intellectual property indemnity, and wilful misconduct
Codevertex indemnity	Codevertex indemnifies Hadia against third-party claims that the software infringes intellectual property rights, and against regulatory fines arising from Codevertex's breach of the data processing agreement, within the cap.
Hadia indemnity	Hadia indemnifies Codevertex against claims arising from Hadia's content, its terms with users, and its handling of funds.

Governing law	Laws of Kenya. Disputes are referred to negotiation for 30 days, then to mediation, then to the courts of Kenya.
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SECTION 19

Risks and mitigations

Risk	Impact	Mitigation	Owner
PayHero onboarding exceeds five weeks	Delayed live payments	Documents requested in week 1; reservation-only launch available; delay terms allocate time and cost by cause	Hadia, Codevertex
Regulatory requirement for payment authorisation	Delay or model change	Legal opinion in weeks 2 to 6; capped holds and balances; instant payout option	Hadia
Fraudulent registries	Financial and reputational loss	Verified payout numbers, limits, first-payout hold, monitoring, moderation	Hadia operations
M-Pesa payout failures	Owner dissatisfaction	Automatic reversal and retry, status enquiry, owner notification, exception queue	Codevertex
Inconsistent retailer data	Slower item entry	Manual entry as primary method	Codevertex
Scope change	Schedule slippage	Change control; agreed exclusions	Both
Holiday period availability	Slower decisions	Holidays scheduled; weekly decision reviews	Both
Low cash gifting adoption	Revenue below expectation	Pilots with planners and church networks; conversion analytics	Hadia

SECTION 20

Post-launch support

Service	Terms
Hypercare	30 days after launch; same-day response to payment incidents
Warranty	Defects against this specification corrected at no cost for 90 days
Ongoing support	Dedicated support and maintenance included in the KES 8,000 monthly service fee
Monitoring	Alerts on payout failures, reconciliation variances, error rates and latency
Hosting and domain	Hosting at KES 12,000 per year, billed annually. Domain purchased and renewed by Hadia, or by Codevertex under the optional managed domain service at the registrar's rate.
Handover	Administration guide, finance guide, runbooks, legal pack, registers, architecture documentation
Enhancements	Native applications, Kiswahili and referral programmes quoted on request

SECTION 21

Acceptance

Signature below confirms acceptance of the scope, schedule and price in this document. Changes are subject to the change control process in [Project delivery](#).

Hadia	Codevertex Africa Limited
Name	
Title	
Signature	
Date	

Appendix: Client question index

Each question raised by Hadia, the short answer, and the section that sets out the detail.

Question	Answer	Detail
A. OWNERSHIP AND CODE		
Who owns the Hadia source code?	Hadia, on payment of the build price	Ownership
Who owns the database and data?	Hadia; Codevertex is processor only	Ownership
Will Hadia receive repository access?	Yes, in Hadia's own GitHub organisation from day one	Repository access
What remains Codevertex IP?	The shared platform services and tooling	Ownership
What licence does Hadia receive?	Non-exclusive platform licence for the agreement term	Platform licence
Can Hadia appoint another developer?	Yes, at any time	Appointing another developer
How does Hadia migrate if support stops?	90 days' notice, full export, 20 hours transition support, optional code escrow	Exit and migration
What is handed over on termination?	Repositories, data, backups, documentation, domain	Handover
B. MONEY		
Complete fee structure, including PayHero	All fees in one table	Fee structure
Is commission on gross or net?	On the full gift; the 2.9% processing fee is paid by the guest by default	Commission basis
Can Hadia change the commission?	Yes, at any time, globally or per registry	Commission basis
Can Hadia run 0% promotions?	Yes, scheduled by date range	Commission basis
Payment reversed after payout?	Recovered from the owner's future contributions	Reversals
Who bears fraud losses?	Hadia as operator, unless caused by a Codevertex defect	Reversals
Card chargebacks?	14-day card hold; debited from registry, reserve, then owner	Reversals
C. MONTHLY COSTS		
What is included in KES 8,000?	Treasury subscription, platform services, monitoring, dedicated support	Monthly service fee
Can it increase?	Fixed 12 months; then at most 10% a year with 60 days' notice	Monthly service fee
Volume growth and usage limits?	No per-transaction fee; fair use to 20,000 contributions a month	Monthly service fee

Cost at 1,000, 5,000 and 10,000 contributions?	KES 14,500, 28,500 and 46,000 a month	Cost examples
D. SECURITY		
Independent penetration test?	Recommended before launch at Hadia's cost; findings fixed free	Penetration testing
Who can access production?	Named Codevertex engineers only, key-based, logged	Production access
Can Codevertex staff see customer data?	Only for support, incidents or restores; logged	Staff access
How is privileged access controlled?	GitOps changes, sealed secrets, firewall, network policies, MFA	Production access
Breach notification time?	Within 24 hours of confirmation	Incident notification
Where is data hosted?	Contabo, EU (Germany); backups on Cloudflare R2	Hosting location
Which processors receive data?	Seven, listed with purpose and location	Data processors
E. DISASTER RECOVERY		
What does RPO 24 hours mean?	Payment records are always recoverable; at most one day of registry edits	Recovery objectives
Backup frequency and location?	Daily; on server and offsite in Cloudflare R2	Backup schedule
Restore tests?	Quarterly, first in week 11, reported to Hadia	Restore testing
If the platform goes down?	Restored within 4 hours; funds safe at PayHero	Outage response
F. PAYMENTS		
M-Pesa or PayHero unavailable?	Alternative rails offered; payouts queue and resume	Failure scenarios
Payout fails?	Funds return to registry and retry automatically	Payout controls
Incorrect M-Pesa number?	Prevented by OTP and name check; reversal if needed	Payout controls
Freeze or manually trigger a payout?	Yes, both	Payout controls
Owner disputes a payout?	Dashboard dispute, 2 working day response	Payout controls
Money when a registry is cancelled?	Refunded to each paying number	Cancellation
G. PRODUCT		
Purchase confirmation, reserved items, manual release	Purchase link, status filters, owner release	Lifecycle rules
Transfer, multiple beneficiaries, date change	Transfer with OTP; one beneficiary; date changes allowed	Lifecycle rules

Expiry and reopening	Closes 30 days after event; reopen within 90 days	Lifecycle rules
H. INTERNATIONAL		
Cards, countries, currencies, FX, fees	Visa and Mastercard worldwide; mobile money in Kenya, Uganda, Tanzania, Rwanda, Burundi, Ethiopia, DR Congo, Zambia, Malawi, Mozambique, Botswana, South Africa, Nigeria and Ghana; KES registries; no Hadia fee to guests	Supported countries
I. CONTRACT		
Termination rights, notice, fee after termination	60 days after initial term; for cause on 30 days' cure; fee ends with notice	Contract terms
Liability cap and indemnity	12 months' fees; IP and data protection indemnity	Contract terms
Missed deadline, PayHero delay, prolonged downtime	Codevertex delay: 2% weekly credit to 10%. Hadia delay: extension plus standby cost. Force majeure: extension, no charges. Downtime: termination right after 72 hours	Contract terms